
Tutorial 2: Solutions

Foundations of Cryptography

Computer Security

BCS2420

Part A

1. Which of the following best describes the advantage of **public-key** (asymmetric) encryption compared to **symmetric encryption**?

Correct Answer: C. It simplifies secure key distribution

Explanation:

- Symmetric encryption often requires a pre-shared key (both sides must have the same secret key).
- Public-key systems allow one to publish a public key while keeping a private key secret, making key distribution more convenient.

2. Which cryptographic property ensures that a given hash output does not reveal any information about the original message?

Correct Answer: B. One-way property

Explanation:

- One-way means it should be infeasible to invert the hash.
- Collision resistance (A) is about finding two inputs with the same hash.
- Second-preimage resistance (D) is about creating a second input matching an existing hash.
- Key encapsulation (C) is unrelated to standard hash properties.

3. In a chosen-ciphertext attack model, the adversary can...

Correct Answer: B. Submit ciphertexts to a decryption oracle and observe the results

Explanation:

- Ciphertext-only: The adversary only sees ciphertext.
- Known-plaintext: The adversary knows some plaintext/ciphertext pairs.
- Chosen-plaintext: The adversary can encrypt plaintext of their choosing.
- **Chosen-ciphertext:** The adversary can *decrypt* chosen ciphertexts and analyze responses.

4. Which statement about the *Vernam cipher* is **correct**?

Correct Answer: D. It can provide perfect secrecy only if the key is truly random and used exactly once.

Explanation:

- Also known as the one-time pad.
- Reusing the key breaks the perfect secrecy property.

5. **In a hybrid encryption scheme, typically...**

Correct Answer: C. A symmetric key is randomly generated for data encryption, then encrypted with a public key.

Explanation:

- Hybrid encryption uses a symmetric key for efficiency. This key is then protected (encrypted) with the recipient's asymmetric public key.

6. **Which of the following is an *active* adversary action?**

Correct Answer: C. Injecting malicious packets into a conversation

Explanation:

- An active adversary modifies or injects data.
- A passive adversary only observes or eavesdrops.

7. **Collision resistance in a hash function means...**

Correct Answer: A. Finding two distinct inputs with the same hash is computationally infeasible.

Explanation:

- "Collision" = different inputs same hash.
- Being hard/impractical to find any two inputs with the same digest is collision resistance.

8. **Which block cipher mode directly concatenates each plaintext block with the preceding ciphertext block (via XOR) before encryption?**

Correct Answer: B. CBC

Explanation:

- CBC (Cipher Block Chaining) mode XORs each plaintext block with the previous block's ciphertext before

encrypting.

- ECB (A) encrypts blocks independently; CTR (C) uses counters; OFB (D) generates keystream blocks.

9. **Digital signatures aim to provide which of the following?**

Correct Answer: B. Non-repudiation, integrity, and origin authentication

Explanation:

- Digital signatures help prove who signed it (origin auth), ensure the message is unaltered (integrity), and prevent the signer from denying having signed (non-repudiation).

10. **A 56-bit key space (as in original DES) implies how many possible keys?**

Correct Answer: A. (2^{56})

Explanation:

- A 56-bit key space has (2^{56}) possible keys. Other expressions are incorrect for that bit size.

Part B

1. Comparing Passive vs. Active Adversaries

- **Passive Adversary:** Observes or eavesdrops on communications without modifying or injecting messages.
 - **Example:** A hacker passively sniffing Wi-Fi traffic to capture data (e.g., passwords).
- **Active Adversary:** Can alter messages in transit, inject new ones, or impersonate legitimate users.
 - **Example:** A man-in-the-middle attack where an attacker intercepts and modifies messages between two parties, or an attacker sending forged commands in a network protocol.
- **Why more challenging to defend against active adversaries?**
 - Active adversaries can manipulate data flows, alter messages, and exploit protocols more dynamically. This requires not just confidentiality but also data integrity checks, authentication protocols, and robust handshake mechanisms.

2. Attack Models

- **Ciphertext-Only Attack (COA)**: Adversary only has ciphertext and attempts to recover plaintext or key.
- **Known-Plaintext Attack (KPA)**: Adversary knows some plaintext/ciphertext pairs and tries to recover the key or decrypt future ciphertext.
- **Chosen-Plaintext Attack (CPA)**: Adversary can choose plaintexts to be encrypted and sees corresponding ciphertexts (e.g., can query an encryption oracle).
- **Chosen-Ciphertext Attack (CCA)**: Adversary can submit chosen ciphertexts to a decryption oracle and see the resulting plaintext.
- **Strongest Requirement**: CCA is generally the most demanding for a cryptosystem because the adversary can fully interact with decryption, providing the highest potential for discovering weaknesses.

3. Key Space and Security

- **Key Space**: All possible keys that can be used by a cryptographic algorithm. Larger key space → more brute-force attempts required.
- **Effect of Increasing Key Length**:
 - Makes brute-force attacks exponentially harder, as each additional bit doubles the key space.
- **Other Practical Factors**:
 1. **Implementation robustness**: Side-channel leaks or poor random number generation can reduce effective security.
 2. **Speed of encryption/decryption**: Slower ciphers can be costlier to brute force, but also can hinder legitimate use if too slow.
 3. (Bonus) **Cryptanalytic breakthroughs**: Algorithms themselves may be broken, reducing effective key length in practice.

4. Designing a Secure Symmetric Cipher

- **Core Properties**:
 1. Confusion (makes relationship between key and ciphertext complex).
 2. Diffusion (spreads plaintext information over many parts of ciphertext).
 3. Large key space + no known structural weaknesses (e.g., no easy differential or linear)

cryptanalysis).

- **Two Advanced Cryptanalytic Methods:**

1. **Differential Cryptanalysis:** Looks at how differences in plaintext input affect differences in ciphertext output. A well-designed cipher incorporates strong non-linear components to resist this.
2. **Linear Cryptanalysis:** Exploits statistical biases in linear combinations of plaintext bits and ciphertext bits. Substitution-permutation networks help mitigate these biases.

5. Public-Key Infrastructure (PKI)

- **Certificates:** Signed data structures that bind a public key to a subject's identity (e.g., a domain or an individual). A Certificate Authority (CA) vouches for correctness by signing.

- This relies on the trust placed in the CA.

- **Use in an Organization:**

1. Each employee gets a certificate with their public key.
2. To send secure email, you retrieve the recipient's **public key** (via their certificate).
3. You encrypt (or sign) email accordingly, trusting the CA's verification of the key's ownership.

6. Hash Functions vs. Encryption Functions

- **Hash Functions:**

- One-way transformations, fixed-length output, used for integrity checks.
 - No secret key necessarily; no direct "decryption" process.

- **Encryption Functions:**

- Transform plaintext to ciphertext with a key; reversible by the corresponding decryption function.

- **Why not use a block cipher as a hash:**

- Block ciphers are designed for two-way encryption/decryption.
 - Hashes require an irreversible property. Simply encrypting with a block cipher does **not** guarantee collision resistance or the one-way property at scale.

7. Digital Signatures and Non-Repudiation

- **Non-Repudiation:** The signer cannot deny having signed a message because only they hold the private key used in generating the signature.
- **E-Commerce Example:** When a customer signs an electronic contract or purchase agreement, the signature ensures they cannot later claim they didn't authorize the transaction.

8. Block Cipher Modes

- **Weakness of ECB:** Identical plaintext blocks produce identical ciphertext blocks → patterns leak.
- **CBC Mode:** Each block is XORed with the previous ciphertext block before encryption, hiding repeated plaintext patterns.
- **CTR Mode Use-Case**:** High-speed streaming or parallel encryption. Each block is encrypted by XORing the plaintext with a keystream block generated from a counter. Often used where random access to encrypted data is needed.

9. Hybrid Encryption Scenario

- **Why Hybrid?**
 - Public-key algorithms are computationally intensive for large data (e.g., 1GB).
 - Symmetric algorithms are fast; so you generate a random symmetric key to encrypt the bulk data, then use the recipient's public key only for encrypting the small symmetric key.
- **Steps:**
 1. Generate a random symmetric key k .
 2. Encrypt the 1GB file with k using a fast symmetric cipher (e.g., AES).
 3. Encrypt k with the recipient's public key.
 4. Store/transmit both (encrypted data + encrypted symmetric key).

10. MAC vs. Digital Signature

- **When MAC is more suitable:**
 - In a **closed system** where all participants share a secret key, and the goal is message authentication + integrity (e.g., within a single company's internal API).
 - MACs are typically faster and simpler to implement

than a signature scheme.

- **When a Digital Signature is better:**

- In **open systems** where you need non-repudiation and publicly verifiable proof of authenticity (e.g., a signed contract, public software release).
- A signature does not require sharing a secret key with the verifier.

Part C

1. Brute-Force Feasibility

- **Key space:** 2^{40} .
 - This is 1,099,511,627,776 possible keys (approximately 1.1×10^{12}).
- **Attacker speed:** 10^9 keys/sec.
 - Worst-case time = $\frac{2^{40}}{10^9} \approx \frac{1.1 \times 10^{12}}{10^9} = 1.1 \times 10^3$ seconds = **1100 seconds**.
 - In days $\approx \frac{1100}{86400} \approx 0.0127$ days 0.3hours.
- **Feasibility:** With modern hardware, cracking a 40-bit key might be feasible in under an hour. This is why 40-bit ciphers are no longer considered secure.

2. Key Length Impact

- **128-bit** vs. **256-bit** keys:
 - Key space(128-bit): 2^{128}
 - Key space(256-bit): 2^{256}
 - The ratio: $\frac{2^{256}}{2^{128}} = 2^{128}$.
 - This is an increase of 2^{128} times, which is astronomically larger.
 - **More than just double:** Doubling key length *squares* the size of the key space when going from 128 to 256 bits.

3. Simple Caesar Cipher Computation

- **Ciphertext:** "FODCGR"
- Assume 'A' = 0, 'B' = 1, ..., 'Z' = 25.
- Caesar shift by 3 (encryption shifts forward by 3). To decrypt, shift backward by 3.
- **Steps** (letter by letter):

- F → F's position is 5 (A=0, B=1, C=2, D=3, E=4, F=5).
 - $5 - 3 = 2 \rightarrow C$.
 - O → O's position is 14.
 - $14 - 3 = 11 \rightarrow L$.
 - D → D's position is 3.
 - $3 - 3 = 0 \rightarrow A$.
 - C → C's position is 2.
 - $2 - 3 = -1 \rightarrow$ wrap around: $-1 + 26 = 25 \rightarrow Z$.
 - G → G's position is 6.
 - $6 - 3 = 3 \rightarrow D$.
 - R → R's position is 17.
 - $17 - 3 = 14 \rightarrow O$.
 - **Plaintext** = "CLAZDO" (assuming standard Caesar with shift of 3).
 - **Number of possible shifts:** For the English alphabet, 26 possible shifts (0 to 25).
4. Block Cipher Encryption Rate
- **Block size:** 128 bits
 - **Throughput:** 1 million 10^6 blocks per second
 - In **bits/sec**: $10^6 \times 128 = 1.28 \times 10^8$ bits/sec = 128 Mbps.
 - In **megabytes/sec**: $\frac{128 \text{ Mbps}}{8} = 16 \text{ MB/s}$.
 - **Time for a 10 GB file:**
 - $10 \text{ GB} = 10 \times 1024 \approx 10,240 \text{ MB}$.
 - At 16 MB/s $\rightarrow \approx \frac{10,240}{16} = 640$ seconds 10.7 minutes .
5. Hash Collisions - **n-bit** hash: collisions around $\sqrt{2^n} = 2^{n/2}$.
- For a **128-bit** hash $\rightarrow \sim \sqrt{2^{128}} = 2^{64}$.
 - That's 1.84×10^{19} .
 - **Why sooner if poorly designed?**
 - Structural weaknesses or patterns can reduce actual complexity needed to find collisions below 2^{64} .
6. Public-Key Encryption Overhead
- **Message count:** 10,000 messages, each 1KB.
 - **RSA encryption:** 5ms per message $\rightarrow$ total time = $10,000 \times 0.005 \text{ s} = 50 \text{ s}$.
 - **Why prefer hybrid?**
 - Public-key operations (RSA) are much slower than symmetric. For *large* volumes of data or numerous messages, using a hybrid scheme drastically reduces total encryption time.

- Also reduces computational overhead on the sender/receiver.